

OS LAB 1B TASKS

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24K-0559

TASK#1).

8).

```
new-user@new-user-VirtualBox:~$ nano file.txt
new-user@new-user-VirtualBox:~$ grep -cie 'Java' -e 'Python' -e 'JavaScript' file.txt
7
new-user@new-user-VirtualBox:~$
```

9).

```
new-user@new-user-VirtualBox:~$ nano part3.txt
new-user@new-user-VirtualBox:~$ grep -e 'server' -e 'database' part1.txt part2.txt part3.txt
part1.txt:Many organizations around the world have adopted Linux for their servers and infrastructure. Companies like Google, Amazon, and Facebook run thousands of Linux servers to power their services. The Linux kernel is constantly being improved by thousands of developers worldwide.
part1.txt:Database management is crucial for modern applications. MySQL and PostgreSQL are popular open-source database systems. MySQL is widely used in web applications, while PostgreSQL is known for its advanced features and standards compliance. Database administrators must ensure data integrity and security.
part1.txt:The concept of Cloud Computing has transformed how businesses operate. Cloud providers like AWS, Azure, and Google Cloud offer scalable infrastructure. Organizations can deploy applications without maintaining physical servers. Cloud computing reduces costs and increases flexibility.
part2.txt:Web development continues to evolve with new frameworks and technologies. JavaScript frameworks like React, Angular, and Vue have transformed frontend development. Backend technologies like Node.js, Django, and Ruby on Rails provide server-side functionality.
part2.txt:Open source software powers much of the modern internet. The Apache HTTP Server remains one of the most popular web servers. Nginx has gained popularity for its performance and scalability. Open source communities collaborate to improve software quality.
new-user@new-user-VirtualBox:~$
```

10).

```
new-user@new-user-VirtualBox:~$ grep -l 'encryption' part1.txt part2.txt part3.txt
part1.txt
part3.txt
new-user@new-user-VirtualBox:~$
```

11).

```
new-user@new-user-VirtualBox:~$ grep -Ff keywords.txt file.txt
```

The **Linux** Operating System has become one of the most important technologies in modern computing. **Linux** was created by Linus Torvalds in 1991 as a free and open-source alternative to proprietary operating systems. The name **Linux** is a combination of Linus and Unix, reflecting its Unix-like architecture. Many organizations around the world have adopted **Linux** for their servers and infrastructure. Companies like Google, Amazon, and Facebook run thousands of **Linux** servers to power their services. The **Linux** kernel is constantly being improved by thousands of developers worldwide.

In the world of programming, **Python** has emerged as a dominant language. **Python** is known for its simplicity and readability. Many beginners choose **Python** as their first programming language because of its gentle learning curve. **Python** supports multiple programming paradigms including object-oriented, procedural, and functional programming.

Database management is crucial for modern applications. MySQL and PostgreSQL are popular open-source database systems. MySQL is widely used in web applications, while PostgreSQL is known for its advanced features and standards compliance. Database administrators must ensure data integrity and **security**.

Security is a critical concern in modern computing. Cyber**security** threats continue to evolve and become more sophisticated. Organizations must implement robust **security** measures including firewalls, encryption, and access controls. Regular **security** audits help identify vulnerabilities.

The Internet of Things (IoT) connects everyday devices to the internet. Smart homes use IoT devices for automation and convenience. Industrial IoT improves manufacturing efficiency through sensor data. However, IoT **security** remains a significant challenge.

System administration requires diverse skills and knowledge. **Linux** administrators manage user accounts, permissions, and services. Shell scripting automates routine tasks. Monitoring tools help maintain system health and performance. Authentication and authorization protect system resources. OAuth enables delegated access to services. Single sign-on simplifies user experience. Multi-factor authentication adds **security** layers.

Scripting languages automate system tasks. Bash scripts execute command sequences. **Python** scripts handle complex logic. PowerShell manages Windows environments.

Compliance requirements govern many industries. Audits verify adherence to standards. Documentation demonstrates compliance efforts. Penalties motivate **security** investments.

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Testing frameworks verify code correctness. JUnit supports Java unit testing. pytest provides **Python** testing features. Jest enables JavaScript testing.

Documentation generators create reference materials. Javadoc produces API documentation. Sphinx builds **Python** project docs. JSDoc documents JavaScript code.

Threat intelligence informs **security** decisions. Indicators of compromise reveal attacks. Sharing communities distribute warnings. Automated feeds enable rapid response.

Vulnerability management identifies and addresses weaknesses. Scanning discovers known vulnerabilities. Prioritization focuses on exploitability. Remediation closes **security** gaps.

Incident response handles **security** breaches. Preparation establishes procedures and tools. Detection identifies potential incidents. Containment limits damage spread.

Forensics investigates **security** events. Evidence preservation maintains integrity. Timeline reconstruction establishes sequences. Attribution identifies responsible parties.

Penetration testing evaluates **security** controls. Authorized attacks reveal vulnerabilities. Reports document findings and recommendations. Retesting verifies remediation effectiveness.

Compliance frameworks provide **security** guidance. NIST Cyber**security** Framework organizes activities. ISO 27001 specifies management requirements. SOC 2 addresses service organization controls.

The **Linux** command line offers powerful text processing capabilities. Grep searches for patterns in files. Sed performs stream editing. Awk processes structured text.

12).

```
? stream editing. Awk processes structured text.
new-user@new-user-VirtualBox:~$ grep '^The' file.txt
The Linux Operating System has become one of the most important technologies in modern comp
The concept of Cloud Computing has transformed how businesses operate. Cloud providers like AWS, Azure, and Google Cloud
offer scalable infrastructure. Organizations can deploy applications without maintaining physical servers. Cloud comput
ing reduces costs and increases flexibility.
The Internet of Things (IoT) connects everyday devices to the internet. Smart homes use IoT devices for automation and c
onvenience. Industrial IoT improves manufacturing efficiency through sensor data. However, IoT security remains a signif
icant challenge.
The future of technology holds exciting possibilities. Quantum computing promises to solve complex problems beyond class
ical computers. Edge computing brings processing closer to data sources. 5G networks enable faster and more reliable con
nectivity.
The global technology industry faces various challenges. Talent shortages affect many specialized areas. Supply chain di
ruptions impact hardware availability. Geopolitical factors influence technology trade and cooperation.
The role of technology in healthcare has expanded dramatically. Electronic health records improve patient care coordinat
ion. Telemedicine enables remote consultations. Medical devices incorporate advanced sensors and connectivity.
The technology sector continues to evolve rapidly. Adaptability and continuous learning are essential for professionals.
New paradigms emerge as existing technologies mature. The intersection of different fields creates novel opportunities.
The command line remains a powerful tool. Shell proficiency increases productivity. Pipelines combine simple commands. A
liases reduce typing for common operations.
The Linux command line offers powerful text processing capabilities. Grep searches for patterns in files. Sed performs s
tream editing. Awk processes structured text.
The technology landscape continues to expand and diversify. Professionals must continually update their skills. Foundati
onal knowledge enables adaptation to new developments. Understanding principles transcends specific tools and platforms.
new-user@new-user-VirtualBox:~$
```

13).

```
stream editing. Awk processes structured text.
The technology landscape continues to expand and diversify. Professionals
onal knowledge enables adaptation to new developments. Understanding prin
new-user@new-user-VirtualBox:~$ grep -i 'data$' file.txt
new-user@new-user-VirtualBox:~$
```

14).

```
new-user@new-user-VirtualBox:~$ grep -Ri "artificialintelligence" tech_articles/
grep: tech_articles/: No such file or directory
```

15).

```
new-user@new-user-VirtualBox:~$ grep -E "AI|Machine Learning|Neural Networks" file.txt
Artificial Intelligence and Machine Learning are driving innovation across industries. Machine learning algorithms can a
nalyze vast amounts of data to discover patterns. Deep learning, a subset of machine learning, uses neural networks with
multiple layers. AI applications range from image recognition to natural language processing.
Ethical considerations in technology are increasingly important. Data privacy regulations like GDPR protect user informa
tion. AI ethics address bias and fairness in algorithms. Technology professionals must consider the societal impact of t
heir work.
new-user@new-user-VirtualBox:~$
```

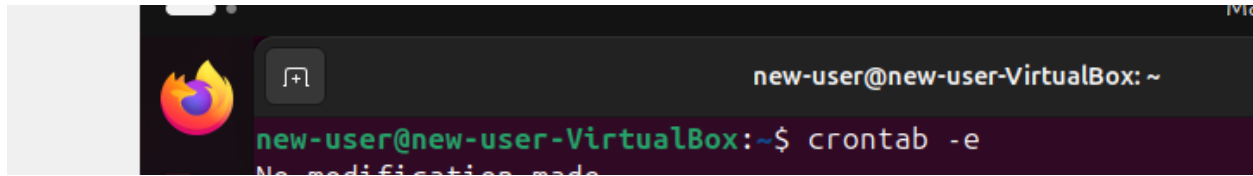
16).

```
new-user@new-user-VirtualBox:~$ grep -q 'blockchain' file.txt && echo "Match Found." || echo "No Match
Found."
Match Found.
new-user@new-user-VirtualBox:~$
```

TASK#2).

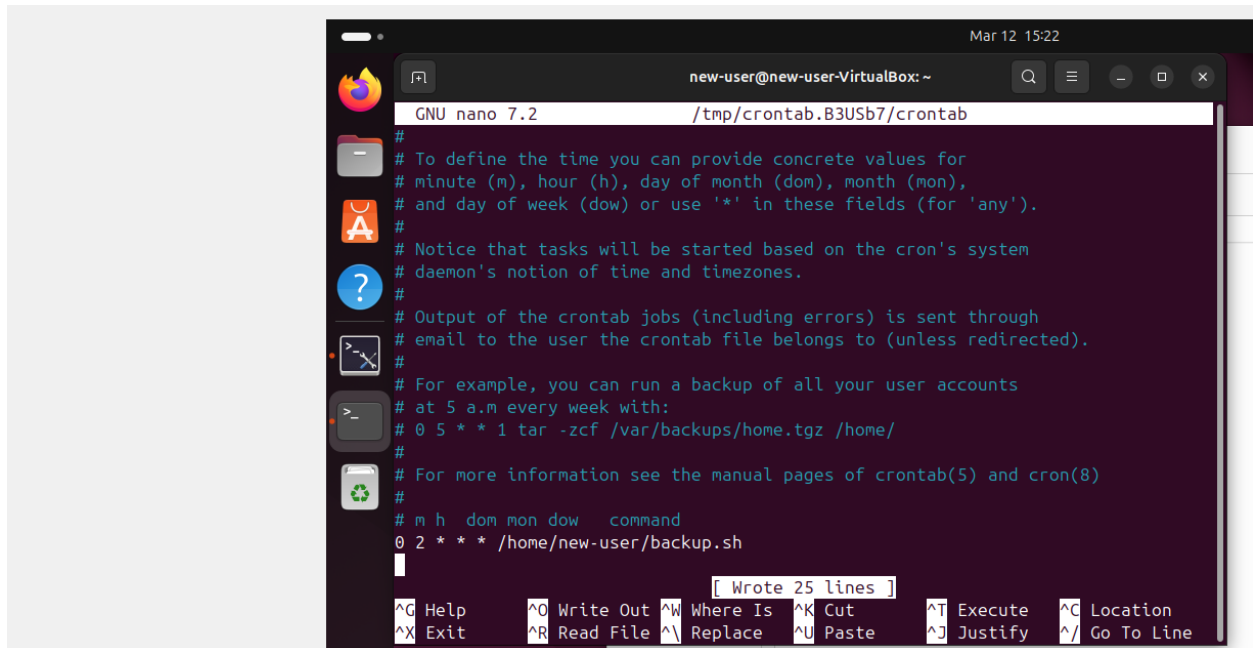
- 1. ps – It is used to display a snapshot/screenshot of currently running processes along with their process PID, status, and resource usage.**
- 2. top – It Shows a dynamic, real-time view of all the running processes and system resource usage.**
- 3. kill – Sends a signal (usually to terminate) to a process using its PID.**
- 4. nice – Starts a process with a specified priority (niceness) to influence CPU scheduling.**
- 5. htop – An interactive, colorized process viewer that allows scrolling, sorting, and managing processes.**

TASK#3).



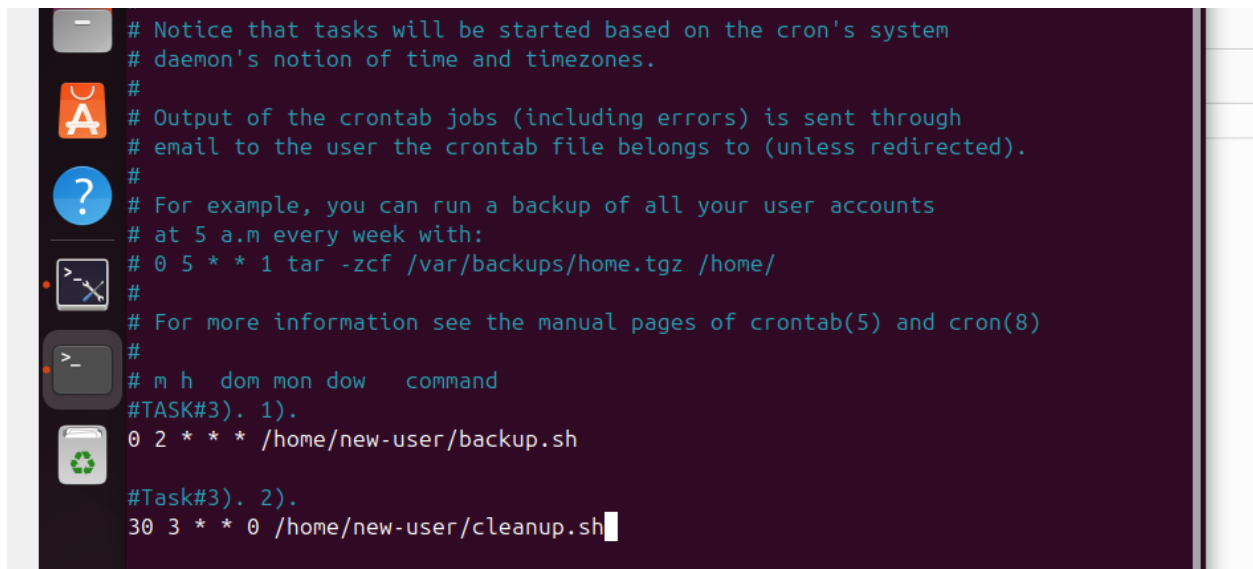
```
new-user@new-user-VirtualBox: ~  
new-user@new-user-VirtualBox:~$ crontab -e  
No modification made
```

1).



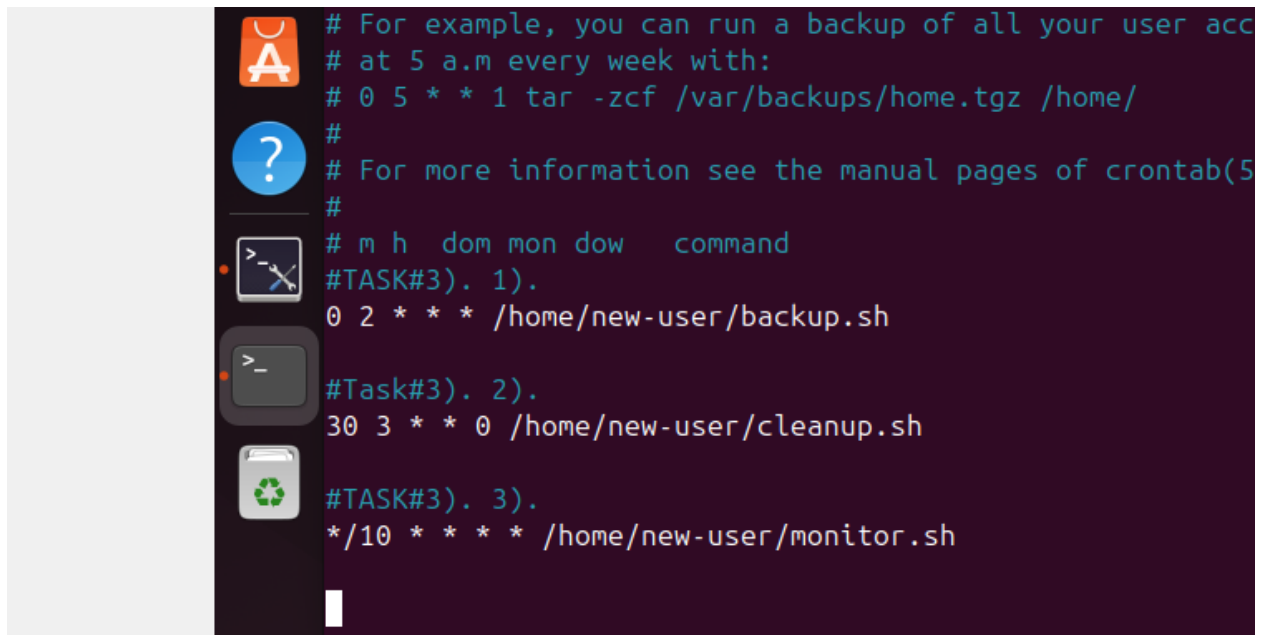
```
GNU nano 7.2 /tmp/crontab.B3USb7/crontab  
#  
# To define the time you can provide concrete values for  
# minute (m), hour (h), day of month (dom), month (mon),  
# and day of week (dow) or use '*' in these fields (for 'any').  
#  
# Notice that tasks will be started based on the cron's system  
# daemon's notion of time and timezones.  
#  
# Output of the crontab jobs (including errors) is sent through  
# email to the user the crontab file belongs to (unless redirected).  
#  
# For example, you can run a backup of all your user accounts  
# at 5 a.m every week with:  
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/  
#  
# For more information see the manual pages of crontab(5) and cron(8)  
#  
# m h dom mon dow  command  
0 2 * * * /home/new-user/backup.sh  
[ Wrote 25 lines ]  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

2).



```
# Notice that tasks will be started based on the cron's system  
# daemon's notion of time and timezones.  
#  
# Output of the crontab jobs (including errors) is sent through  
# email to the user the crontab file belongs to (unless redirected).  
#  
# For example, you can run a backup of all your user accounts  
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# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/  
#  
# For more information see the manual pages of crontab(5) and cron(8)  
#  
# m h dom mon dow  command  
#TASK#3). 1).  
0 2 * * * /home/new-user/backup.sh  
  
#Task#3). 2).  
30 3 * * * 0 /home/new-user/cleanup.sh
```

3).

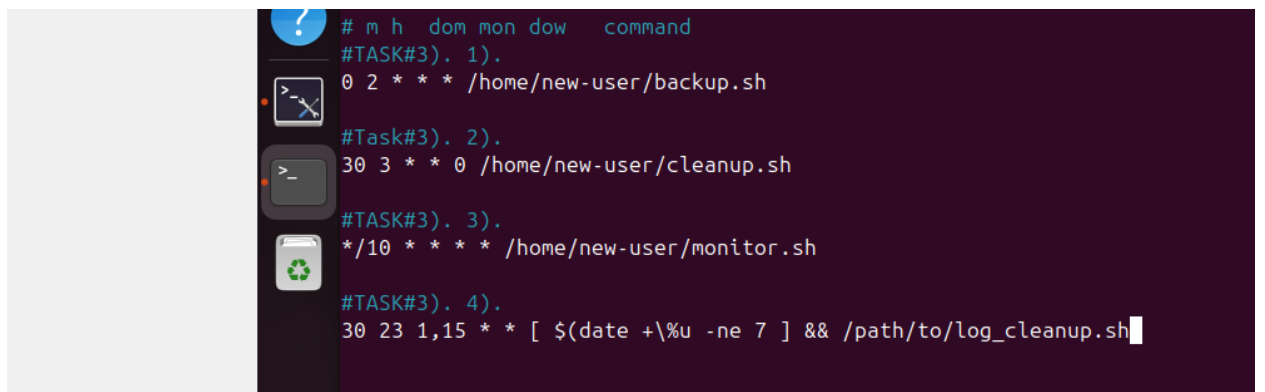
A terminal window with a dark purple background and a light gray sidebar on the left. The sidebar contains five icons: an orange 'A' logo, a blue circle with a white question mark, a terminal icon with a red cursor, a terminal icon with a red cursor, and a green recycling symbol. The terminal text is as follows:

```
# For example, you can run a backup of all your user acc
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
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#
# m h dom mon dow    command
#TASK#3). 1).
0 2 * * * /home/new-user/backup.sh

#Task#3). 2).
30 3 * * 0 /home/new-user/cleanup.sh

#TASK#3). 3).
*/10 * * * * /home/new-user/monitor.sh
```

4).

A terminal window with a dark purple background and a light gray sidebar on the left. The sidebar contains five icons: a blue circle with a white question mark, a terminal icon with a red cursor, a terminal icon with a red cursor, a terminal icon with a red cursor, and a green recycling symbol. The terminal text is as follows:

```
# m h dom mon dow    command
#TASK#3). 1).
0 2 * * * /home/new-user/backup.sh

#Task#3). 2).
30 3 * * 0 /home/new-user/cleanup.sh

#TASK#3). 3).
*/10 * * * * /home/new-user/monitor.sh

#TASK#3). 4).
30 23 1,15 * * [ $(date +%u -ne 7 ) && /path/to/log_cleanup.sh
```